A Spectral Drift–Diffusion Stability Criterion for Linear Stochastic Dynamical Systems with Applications to Neural ODEs

Devang Verma

Abstract

Continuous-time neural dynamical systems, including Neural Ordinary Differential Equations (Neural ODEs) and Neural Stochastic Differential Equations (Neural SDEs), are increasingly deployed in noisy and uncertain environments. Ensuring stability under multiplicative stochastic perturbations is therefore critical. Classical mean-square exponential stability conditions rely on scalar contraction–diffusion inequalities that treat drift and diffusion isotropically and may be overly conservative. In this work, we derive a spectral sufficient condition for mean-square exponential stability of linear multiplicative stochastic systems. The condition is expressed in terms of the maximal eigenvalue of a combined drift–diffusion matrix and corresponds to the identity Lyapunov matrix case of classical LMI-based stability theory. We prove that this spectral criterion strictly generalizes standard scalar contraction bounds and demonstrate this via an explicit counterexample. Finally, we extend the framework to nonlinear stochastic neural ODEs through Jacobian linearization, providing a practical spectral stability test for neural dynamical systems.

1 Introduction

1.1 Motivation

Continuous-time neural dynamical systems have become foundational tools in modern machine learning and computational neuroscience. Neural Ordinary Differential Equations (Neural ODEs) model representation dynamics through continuous flows, enabling memory-efficient training and principled dynamical analysis. Their stochastic counterparts, Neural Stochastic Differential Equations (Neural SDEs), incorporate noise to model uncertainty, irregular time series, and diffusion-driven processes. In practical deployments, neural dynamical systems are inevitably subject to stochastic perturbations. These perturbations may arise from environmental uncertainty, sensor noise, discretization effects, or intrinsic stochasticity in biological neural circuits. As a result, understanding the stability of continuous-time neural systems under multiplicative noise is essential for ensuring reliable performance, interpretability, and robustness. While deterministic stability theory is well developed, stochastic stability introduces additional complexity. In particular, multiplicative noise interacts directionally with the drift dynamics, potentially destabilizing otherwise contracting systems. Classical mean-square exponential stability results typically rely on scalar contraction–diffusion tradeoffs that bound drift and diffusion effects isotropically. Such bounds, while sufficient, can be overly conservative and fail to capture directional structure inherent in neural architectures. This motivates the development of refined stability criteria that explicitly account for the spectral interaction between drift and diffusion components.

1.2 Problem Statement

We consider linear stochastic dynamical systems of the form

$$dz = Az\,dt + \sum_{i=1}^{m} B_i z\,dW_t^{(i)}, \tag{1}$$

where $A, B_i \in \mathbb{R}^{d\times d}$ and $W_t^{(i)}$ are independent standard Brownian motions. This formulation naturally arises as:

- Local linearizations of nonlinear stochastic neural ODEs,
- Continuous-time limits of stochastic recurrent neural networks,
- Linear approximations of diffusion-regularized latent dynamics.

The central question we address is:

> Under what explicit spectral conditions on A and $\{B_i\}$ is the equilibrium $z = 0$ mean-square exponentially stable?

More specifically, we seek a condition that:

- Is easy to compute,
- Avoids solving full Lyapunov matrix inequalities,
- Improves upon classical scalar contraction bounds,
- Extends naturally to nonlinear neural dynamical systems.

1.3 Contributions

The main contributions of this work are as follows:

1. We derive a spectral sufficient condition for mean-square exponential stability based on the maximal eigenvalue of a combined drift–diffusion matrix.
2. We formally establish that this condition corresponds to the identity Lyapunov metric case of classical LMI-based stochastic stability theory.
3. We prove that the proposed spectral criterion strictly generalizes classical scalar contraction–diffusion bounds and provide an explicit counterexample demonstrating strict improvement.
4. We extend the framework to nonlinear stochastic neural ODEs via Jacobian linearization, yielding a practical spectral stability test for neural dynamical systems.

Collectively, these results provide a simple yet powerful stability characterization that captures directional drift–diffusion interactions and offers practical implications for the design and regularization of stochastic neural architectures.

2 Preliminaries and Problem Setup

2.1 Notation

Throughout this paper, $\|\cdot\|$ denotes the Euclidean norm on $\mathbb{R}^d$, and for a matrix $M \in \mathbb{R}^{d\times d}$, $\|M\|$ denotes the induced operator norm. The notation M^T denotes the transpose of M. For symmetric matrices $M, N \in \mathbb{R}^{d\times d}$, we write

$$M \preceq N$$

if $N - M$ is positive semidefinite, and

$$M \prec N$$

if $N - M$ is positive definite. The maximal eigenvalue of a symmetric matrix M is denoted by

$$\lambda_{\max}(M).$$

Expectation with respect to the underlying probability measure is denoted by $\mathbb{E}[\cdot]$.

2.2 Linear Stochastic Dynamical Systems

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space equipped with a filtration $\{\mathcal{F}_t\}_{t\geq 0}$ satisfying the usual conditions. Let $\{W_t^{(i)}\}_{i=1}^m$ be independent standard Brownian motions adapted to this filtration. We consider linear multiplicative stochastic differential equations of the form

$$dz(t) = Az(t)\,dt + \sum_{i=1}^{m} B_i z(t)\,dW_t^{(i)}, \tag{2}$$

where:

- $A \in \mathbb{R}^{d\times d}$ is the drift matrix,
- $B_i \in \mathbb{R}^{d\times d}$ are diffusion matrices,
- $z(t) \in \mathbb{R}^d$ is the state process.

We assume that the coefficients are globally Lipschitz and of linear growth, ensuring existence and uniqueness of strong solutions to (2). This class of systems arises naturally as:

- Local linearizations of nonlinear stochastic neural ODEs,
- Continuous-time limits of stochastic recurrent neural networks,
- Latent linear dynamics under multiplicative noise.

2.3 Mean-Square Exponential Stability

We now formalize the notion of stochastic stability used throughout this paper.

Definition 1 (Mean-Square Exponential Stability)**.** The equilibrium $z = 0$ of (2) is said to be *mean-square exponentially stable* if there exist constants $C > 0$ and $\beta > 0$ such that for all initial conditions $z(0)$,

$$\mathbb{E}\|z(t)\|^2 \leq Ce^{-\beta t}\|z(0)\|^2 \quad \text{for all } t \geq 0.$$

This notion captures exponential decay of the second moment of the state and is the stochastic analogue of deterministic exponential stability. Mean-square stability is weaker than almost-sure exponential stability but stronger than mere boundedness in expectation. For linear systems of the form (2), it provides a tractable and practically meaningful notion of robustness under multiplicative noise.

2.4 Lyapunov Approach to Mean-Square Stability

A classical approach to analyzing mean-square stability is via quadratic Lyapunov functions of the form

$$V(z) = z^T P z,$$

where $P \succ 0$ is a symmetric positive definite matrix. Applying Itô's formula to $V(z(t))$ yields

$$dV = \left(z^T (A^T P + PA) z + \sum_{i=1}^{m} z^T B_i^T P B_i z \right) dt + \text{martingale terms.}$$

Taking expectations eliminates martingale terms and gives

$$\frac{d}{dt}\mathbb{E}[V(z)] = \mathbb{E}\left[z^T \left(A^T P + PA + \sum_{i=1}^{m} B_i^T P B_i \right) z \right].$$

A sufficient condition for mean-square exponential stability is therefore the Linear Matrix Inequality (LMI)

$$A^T P + PA + \sum_{i=1}^{m} B_i^T P B_i \prec 0. \tag{3}$$

While this condition is powerful and, in fact, necessary and sufficient for linear systems, solving (3) requires semidefinite programming and does not yield a simple closed-form spectral characterization.

2.5 Classical Scalar Contraction–Diffusion Condition

A commonly used sufficient condition in neural and control literature imposes isotropic bounds on drift and diffusion separately. Assume there exist constants $\alpha > 0$ and $\gamma > 0$ such that

$$A + A^T \preceq -2\alpha I, \tag{4}$$

$$\sum_{i=1}^{m} B_i^T B_i \preceq \gamma^2 I. \tag{5}$$

Then

$$A + A^T + \sum_{i=1}^{m} B_i^T B_i \preceq (-2\alpha + \gamma^2) I.$$

If

$$2\alpha > \gamma^2,$$

the right-hand side is negative definite, implying mean-square exponential stability. This scalar contraction–diffusion tradeoff treats all directions uniformly and ignores directional alignment between drift and diffusion matrices. As a result, it can be conservative, particularly in anisotropic systems.

2.6 Objective of This Work

The goal of this paper is to derive and analyze a refined spectral stability criterion that:

- Directly characterizes stability through eigenvalues,
- Avoids solving general LMIs,
- Improves upon scalar contraction bounds,
- Extends naturally to nonlinear stochastic neural ODEs.

In the next section, we develop such a spectral drift–diffusion condition and analyze its properties.

3 Spectral Drift–Diffusion Stability Result

We now derive the central theoretical result of this paper: a spectral sufficient condition for mean-square exponential stability of linear multiplicative stochastic systems.

3.1 Generator Computation

Consider the linear stochastic system

$$dz(t) = Az(t)\,dt + \sum_{i=1}^{m} B_i z(t)\,dW_t^{(i)}. \tag{6}$$

Let us define the quadratic Lyapunov function

$$V(z) = z^T z = \|z\|^2. \tag{7}$$

Since $V \in C^2(\mathbb{R}^d)$, Itô's formula applies. Recall that for an SDE of the form

$$dz = f(z)\,dt + \sum_{i=1}^{m} g_i(z)\,dW_t^{(i)},$$

Itô's formula gives

$$dV(z) = \nabla V(z)^T f(z)\,dt + \frac{1}{2}\sum_{i=1}^{m} g_i(z)^T \nabla^2 V(z) g_i(z)\,dt + \sum_{i=1}^{m} \nabla V(z)^T g_i(z)\,dW_t^{(i)}.$$

In our case:

$$f(z) = Az, \quad g_i(z) = B_i z.$$

We compute derivatives of V:

$$\nabla V(z) = 2z, \quad \nabla^2 V(z) = 2I.$$

Substituting into Itô's formula:

$$dV = 2z^T Az\,dt + \sum_{i=1}^{m} (B_i z)^T (2I)(B_i z)\frac{1}{2}dt + 2\sum_{i=1}^{m} z^T B_i z\,dW_t^{(i)}.$$

Simplifying:

$$dV = \left(2z^T Az + \sum_{i=1}^{m} z^T B_i^T B_i z\right) dt + 2\sum_{i=1}^{m} z^T B_i z\,dW_t^{(i)}.$$

3.2 Taking Expectations

The stochastic integral term is a martingale with zero expectation. Therefore,

$$\frac{d}{dt}\mathbb{E}[V(z(t))] = \mathbb{E}\left[2z^T Az + \sum_{i=1}^{m} z^T B_i^T B_i z\right].$$

Note that

$$2z^T Az = z^T(A + A^T)z.$$

Hence,

$$\frac{d}{dt}\mathbb{E}\|z(t)\|^2 = \mathbb{E}\left[z^T\left(A + A^T + \sum_{i=1}^{m} B_i^T B_i\right)z\right]. \tag{8}$$

Define the symmetric matrix

$$M := A + A^T + \sum_{i=1}^{m} B_i^T B_i. \tag{9}$$

Equation (8) becomes

$$\frac{d}{dt}\mathbb{E}\|z(t)\|^2 = \mathbb{E}[z^T Mz]. \tag{10}$$

3.3 Spectral Estimate

Since M is symmetric, it admits an orthonormal eigenbasis. For any $z \in \mathbb{R}^d$,

$$z^T Mz \leq \lambda_{\max}(M)\|z\|^2.$$

Therefore,

$$\frac{d}{dt}\mathbb{E}\|z(t)\|^2 \leq \lambda_{\max}(M)\mathbb{E}\|z(t)\|^2. \tag{11}$$

3.4 Main Stability Theorem

Theorem 1 (Spectral Drift–Diffusion Stability)**.** Let $M = A + A^T + \sum_{i=1}^{m} B_i^T B_i$. If

$$\lambda_{\max}(M) < 0,$$

then the equilibrium $z = 0$ of (6) is mean-square exponentially stable. Moreover,

$$\mathbb{E}\|z(t)\|^2 \leq e^{\lambda_{\max}(M)t}\|z(0)\|^2.$$

Proof. Let $\lambda_{\max}(M) = -\varepsilon$ with $\varepsilon > 0$.

From (11),

$$\frac{d}{dt}\mathbb{E}\|z(t)\|^2 \leq -\varepsilon\mathbb{E}\|z(t)\|^2.$$

Define

$$y(t) := \mathbb{E}\|z(t)\|^2.$$

Then

$$y'(t) \leq -\varepsilon y(t).$$

By Gronwall's inequality,

$$y(t) \leq e^{-\varepsilon t} y(0).$$

Since $y(0) = \|z(0)\|^2$, the result follows. □

3.5 Interpretation

The matrix

$$M = A + A^T + \sum B_i^T B_i$$

captures the combined effect of:

- Deterministic contraction (through $A + A^T$),
- Stochastic diffusion energy (through $B_i^T B_i$).

The largest eigenvalue of M represents the worst-case growth direction of the second moment. Stability requires that even in the most dangerous direction, contraction dominates diffusion.

3.6 Relation to Lyapunov Matrix Inequality

The general Lyapunov LMI condition requires finding $P \succ 0$ such that

$$A^T P + PA + \sum B_i^T P B_i \prec 0.$$

Our condition corresponds to the special case $P = I$. Thus:

Remark 1. The spectral condition is the identity-metric instance of the general quadratic Lyapunov criterion. While more restrictive than the full LMI condition, it yields a closed-form eigenvalue test.

3.7 Sharpness of the Criterion

The bound is exact in the sense that the decay rate of the second moment satisfies

$$\limsup_{t \to \infty} \frac{1}{t} \log \mathbb{E}\|z(t)\|^2 \leq \lambda_{\max}(M).$$

Thus $\lambda_{\max}(M)$ directly controls the exponential growth or decay rate of the second moment.

4 Comparison with Classical Contraction–Diffusion Criteria

In this section, we compare the proposed spectral stability condition with the classical scalar contraction–diffusion bound commonly used in stochastic control and neural stability analysis.

4.1 Classical Scalar Contraction–Diffusion Condition

A widely used sufficient condition for mean-square exponential stability assumes the existence of constants $\alpha > 0$ and $\gamma > 0$ such that

$$A + A^T \preceq -2\alpha I, \tag{12}$$

$$\sum_{i=1}^{m} B_i^T B_i \preceq \gamma^2 I, \tag{13}$$

with the additional requirement

$$2\alpha > \gamma^2. \tag{14}$$

Under these assumptions, one obtains

$$A + A^T + \sum_{i=1}^{m} B_i^T B_i \preceq (-2\alpha + \gamma^2) I.$$

If $2\alpha > \gamma^2$, the right-hand side is negative definite, implying mean-square exponential stability via the identity Lyapunov function. This condition treats drift and diffusion isotropically, bounding each separately by scalar multiples of the identity.

4.2 Relation to the Spectral Criterion

Recall that our spectral stability condition requires

$$\lambda_{\max}\left(A + A^T + \sum_{i=1}^{m} B_i^T B_i\right) < 0.$$

Define

$$M := A + A^T + \sum_{i=1}^{m} B_i^T B_i.$$

We now establish the precise relationship between the two criteria.

Proposition 1. If the classical scalar contraction–diffusion conditions (12)–(14) hold, then the spectral condition holds.

Proof. From (12) and (13), we have

$$M = A + A^T + \sum B_i^T B_i \preceq (-2\alpha + \gamma^2) I.$$

If $2\alpha > \gamma^2$, then $-2\alpha + \gamma^2 < 0$, so

$$M \prec 0.$$

Therefore,

$$\lambda_{\max}(M) < 0.$$

Hence, the spectral condition holds. □

4.3 Strict Generalization

The converse does not hold in general. That is, the spectral condition may hold even when the scalar contraction–diffusion bounds fail.

Theorem 2 (Strict Generalization)**.** There exist matrices A and B_i such that

$$\lambda_{\max}\left(A + A^T + \sum B_i^T B_i\right) < 0,$$

but no scalars $\alpha, \gamma > 0$ satisfy (12)–(14).

Proof. Consider the two-dimensional example

$$A = \begin{pmatrix} -5 & 0 \\ 0 & -0.5 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then

$$A + A^T = \begin{pmatrix} -10 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B^T B = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus

$$M = \begin{pmatrix} -6 & 0 \\ 0 & -1 \end{pmatrix}.$$

Since $\lambda_{\max}(M) = -1 < 0$, the spectral condition holds. However, for the scalar contraction condition to hold, we require

$$A + A^T \preceq -2\alpha I.$$

The smallest eigenvalue of $-(A + A^T)$ is 1, so we must take $\alpha \leq 0.5$. Then

$$\sum B_i^T B_i \preceq \gamma^2 I$$

requires $\gamma^2 \geq 4$. But then

$$2\alpha - \gamma^2 \leq 1 - 4 = -3 < 0,$$

violating the scalar tradeoff condition. Hence, the scalar contraction criterion fails while the spectral condition holds. □

4.4 Geometric Interpretation

The classical scalar condition enforces uniform contraction across all directions and bounds diffusion uniformly as well. In contrast, the spectral condition accounts for directional alignment between drift and diffusion. In particular, diffusion that acts primarily in strongly contracting directions may not destabilize the system. The scalar criterion cannot capture this anisotropic interaction and therefore may be conservative.

4.5 Summary of Logical Implications

We conclude:

$$\text{Scalar Contraction Condition} \quad \Longrightarrow \quad \text{Spectral Condition},$$

but

$$\text{Spectral Condition} \quad \not\Longrightarrow \quad \text{Scalar Condition}.$$

Thus, the spectral drift–diffusion criterion strictly generalizes the classical scalar contraction–diffusion bound.

5 Counterexample Illustrating Strict Improvement

We now construct a concrete two-dimensional example demonstrating that the spectral drift–diffusion condition certifies mean-square stability in situations where the classical scalar contraction–diffusion bound fails.

5.1 System Definition

Consider the linear stochastic system

$$dz = Az\,dt + Bz\,dW_t,$$

where

$$A = \begin{pmatrix} -5 & 0 \\ 0 & -0.5 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

This system exhibits strong contraction in the first coordinate and weak contraction in the second, while stochastic forcing acts only in the first coordinate.

5.2 Step 1: Compute the Spectral Matrix

Recall the spectral drift–diffusion matrix

$$M = A + A^T + B^T B.$$

Since A is diagonal,

$$A + A^T = \begin{pmatrix} -10 & 0 \\ 0 & -1 \end{pmatrix}.$$

Further,

$$B^T B = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix}.$$

Thus,

$$M = \begin{pmatrix} -6 & 0 \\ 0 & -1 \end{pmatrix}.$$

The eigenvalues of M are

$$\lambda_1 = -6, \qquad \lambda_2 = -1.$$

Hence,

$$\lambda_{\max}(M) = -1 < 0.$$

By Theorem 1, the system is mean-square exponentially stable, with decay rate at least 1.

5.3 Step 2: Attempt to Apply the Scalar Contraction Criterion

We now test whether the classical scalar contraction–diffusion condition holds. The drift condition requires

$$A + A^T \preceq -2\alpha I.$$

Since

$$A + A^T = \begin{pmatrix} -10 & 0 \\ 0 & -1 \end{pmatrix},$$

the largest eigenvalue of $A + A^T$ is -1. Therefore, the strongest possible α satisfying

$$A + A^T \preceq -2\alpha I$$

is

$$2\alpha = 1 \quad \Rightarrow \quad \alpha = 0.5.$$

Next, the diffusion bound requires

$$B^T B \preceq \gamma^2 I.$$

Since

$$B^T B = \begin{pmatrix} 4 & 0 \\ 0 & 0 \end{pmatrix},$$

the largest eigenvalue is 4, so we must have

$$\gamma^2 \geq 4.$$

The scalar tradeoff condition requires

$$2\alpha > \gamma^2.$$

But

$$2\alpha = 1, \qquad \gamma^2 \geq 4,$$

so

$$2\alpha - \gamma^2 \leq 1 - 4 = -3 < 0.$$

Thus the scalar contraction–diffusion condition fails.

5.4 Step 3: Analytical Interpretation

The scalar test fails because it treats contraction isotropically. It compares:

$$\text{Worst-case contraction strength} = 1$$

against

$$\text{Worst-case diffusion energy} = 4,$$

without considering that the diffusion acts only in the direction of strong contraction. In contrast, the spectral matrix

$$M = \begin{pmatrix} -6 & 0 \\ 0 & -1 \end{pmatrix}$$

correctly captures directional interaction:

- In the first coordinate, drift contraction -10 is partially offset by diffusion $+4$, yielding net -6.
- In the second coordinate, there is no diffusion and contraction remains -1.

Since both directions remain strictly negative, mean-square exponential stability holds.

5.5 Conclusion of the Example

This example demonstrates:

$$\text{Spectral Condition Holds} \quad \text{while} \quad \text{Scalar Condition Fails.}$$

Therefore, the spectral drift–diffusion criterion strictly generalizes the classical scalar contraction–diffusion bound.

5.6 Kronecker Representation and Generator Spectrum

The covariance equation (**??**) can be written in vectorized form. Using the identity

$$\mathrm{vec}(A\Sigma B) = (B^T \otimes A)\mathrm{vec}(\Sigma),$$

equation (**??**) becomes

$$\frac{d}{dt}\mathrm{vec}(\Sigma) = (I \otimes A + A \otimes I + B \otimes B)\,\mathrm{vec}(\Sigma).$$

Define the generator matrix

$$\mathcal{L} = I \otimes A + A \otimes I + B \otimes B.$$

For this example, direct computation shows that the eigenvalues of $\mathcal{L}$ are

$$\{-6,\ -5.5,\ -5.5,\ -1\}.$$

The maximal eigenvalue is -1. Thus the covariance dynamics decay at rate 1, consistent with the spectral criterion.

5.7 Interpretation

The spectral drift–diffusion matrix M captures the worst-case growth direction of the second moment. The Kronecker generator $\mathcal{L}$ governs full covariance evolution. In this example:

$$\max \operatorname{spec}(\mathcal{L}) = \lambda_{\max}(M).$$

This confirms that the spectral criterion precisely captures the dominant eigenvalue of the second-moment generator.

6 Extension to Nonlinear Stochastic Neural ODEs

We now extend the spectral drift–diffusion stability criterion to nonlinear stochastic dynamical systems, including stochastic neural ODE models.

6.1 Nonlinear Stochastic System

Consider the nonlinear stochastic differential equation

$$dz = f(z)\,dt + \sum_{i=1}^{m} g_i(z)\,dW_t^{(i)}, \tag{15}$$

where:

- $f : \mathbb{R}^d \to \mathbb{R}^d$ is continuously differentiable,
- $g_i : \mathbb{R}^d \to \mathbb{R}^d$ are continuously differentiable,
- $W_t^{(i)}$ are independent Brownian motions.

We assume global Lipschitz continuity to guarantee existence and uniqueness of strong solutions.

6.2 Equilibrium and Linearization

Let z^* be an equilibrium point satisfying

$$f(z^*) = 0, \qquad g_i(z^*) = 0 \quad \text{for all } i.$$

The assumption $g_i(z^*) = 0$ ensures that the equilibrium is not constantly forced by additive noise. Define the perturbation variable

$$y = z - z^*.$$

Expanding f and g_i using first-order Taylor approximation near z^*,

$$f(z^* + y) = J_f(z^*)y + R_f(y),$$

$$g_i(z^* + y) = J_{g_i}(z^*)y + R_{g_i}(y),$$

where:

$$J_f(z^*) = \nabla f(z^*), \qquad J_{g_i}(z^*) = \nabla g_i(z^*),$$

and the remainder terms satisfy

$$\|R_f(y)\| = o(\|y\|), \qquad \|R_{g_i}(y)\| = o(\|y\|).$$

The linearized SDE becomes

$$dy = J_f(z^*)y\,dt + \sum_{i=1}^{m} J_{g_i}(z^*)y\,dW_t^{(i)} + \text{higher-order terms.} \tag{16}$$

6.3 Local Mean-Square Stability Criterion

Define the symmetric matrix

$$M(z) := J_f(z) + J_f(z)^T + \sum_{i=1}^{m} J_{g_i}(z)^T J_{g_i}(z). \tag{17}$$

We now state the nonlinear extension of the spectral criterion.

Theorem 3 (Local Spectral Mean-Square Stability)**.** Suppose there exists a neighborhood $\mathcal{D}$ of z^* such that

$$\sup_{z\in\mathcal{D}} \lambda_{\max}\big(M(z)\big) < 0. \tag{18}$$

Then z^* is locally mean-square exponentially stable for the nonlinear stochastic system (15).

Proof. Consider the Lyapunov function

$$V(y) = \|y\|^2.$$

Applying Itô's formula to (15) expressed in perturbation coordinates yields

$$\frac{d}{dt}\mathbb{E}\|y\|^2 = \mathbb{E}\left[2y^T f(z^*+y) + \sum_{i=1}^{m} \|g_i(z^*+y)\|^2\right].$$

Using first-order expansion,

$$2y^T f(z^*+y) = y^T\big(J_f(z^*) + J_f(z^*)^T\big)y + o(\|y\|^2),$$

and similarly,

$$\|g_i(z^*+y)\|^2 = y^T J_{g_i}(z^*)^T J_{g_i}(z^*)y + o(\|y\|^2).$$

Thus, for sufficiently small $\|y\|$,

$$\frac{d}{dt}\mathbb{E}\|y\|^2 \le \mathbb{E}\left[y^T M(z^*)y\right] + \varepsilon\mathbb{E}\|y\|^2,$$

for arbitrarily small $\varepsilon > 0$. If (18) holds, there exists $\delta > 0$ such that

$$y^T M(z)y \le -\delta\|y\|^2 \quad \text{for } z \in \mathcal{D}.$$

Therefore,

$$\frac{d}{dt}\mathbb{E}\|y\|^2 \leq -\delta\mathbb{E}\|y\|^2$$

for sufficiently small initial conditions. Applying Gronwall's inequality yields

$$\mathbb{E}\|y(t)\|^2 \leq Ce^{-\delta t}\|y(0)\|^2.$$

Hence, z^* is locally mean-square exponentially stable. □

7 Discussion

7.1 Directional Interaction Between Drift and Diffusion

The central insight of the spectral drift–diffusion criterion lies in its directional characterization of stochastic stability. Classical scalar contraction–diffusion bounds treat drift and diffusion isotropically. They require uniform contraction strength across all directions and impose a uniform upper bound on diffusion magnitude. This effectively compares two scalars:

Worst-case contraction strength vs. Worst-case diffusion energy.

Such scalar tests ignore the geometric alignment between the drift and diffusion operators. In contrast, the spectral matrix

$$M = A + A^T + \sum_{i=1}^{m} B_i^T B_i$$

captures directional interaction explicitly. Stability is determined by the maximal eigenvalue of M, which corresponds to the most unstable direction of second-moment growth. In particular:

- Diffusion acting in strongly contracting directions may not destabilize the system.
- Diffusion acting in weakly contracting or expanding directions has greater destabilizing effect.

Thus, the spectral criterion reflects anisotropic structure and can certify stability in systems where scalar bounds fail.

7.2 Why the Spectral Condition is Less Conservative

The scalar contraction condition enforces

$$A + A^T \preceq -2\alpha I, \quad \sum B_i^T B_i \preceq \gamma^2 I,$$

thereby constraining each matrix relative to the identity. This effectively assumes the identity metric is appropriate for measuring contraction in all directions equally. The spectral condition, however, evaluates the combined matrix directly:

$$\lambda_{\max}\left(A + A^T + \sum B_i^T B_i\right) < 0.$$

It does not require drift or diffusion to be bounded individually by multiples of the identity. Instead, it allows cancellation effects between strong contraction and diffusion in aligned directions. As demonstrated by the explicit counterexample, the scalar test may fail even when the true second-moment dynamics decay exponentially.

7.3 Relation to Lyapunov and LMI Approaches

The most general quadratic mean-square stability condition for linear stochastic systems is the existence of a positive definite matrix $P \succ 0$ such that

$$A^T P + PA + \sum_{i=1}^{m} B_i^T P B_i \prec 0.$$

This is a Linear Matrix Inequality (LMI) and provides a necessary and sufficient condition for mean-square exponential stability. The spectral condition derived in this paper corresponds to the special case $P = I$. Thus, it can be viewed as an identity-metric Lyapunov test. While more restrictive than the full LMI condition, the spectral test has several advantages:

- It admits a closed-form eigenvalue characterization.
- It avoids semidefinite programming.
- It provides direct interpretability via eigenstructure.
- It is computationally lightweight for large-scale neural systems.

In neural ODE contexts, where Jacobians are already computed during backpropagation, evaluating the maximal eigenvalue of the spectral matrix may be significantly simpler than solving a full LMI.

7.4 Connection to Second-Moment Generator Spectrum

The covariance dynamics evolve according to

$$\frac{d}{dt}\Sigma = A\Sigma + \Sigma A^T + \sum B_i \Sigma B_i^T.$$

Vectorization reveals a linear generator operator governing second-moment evolution. The spectral condition bounds the maximal eigenvalue of this generator through a quadratic form argument. In certain cases, as shown in the counterexample, the spectral bound precisely matches the true second-moment decay rate.

7.5 Implications for Neural Dynamical Systems

For stochastic neural ODEs, the nonlinear extension reveals that stability depends on the maximal eigenvalue of

$$J_f(z) + J_f(z)^T + \sum_i J_{g_i}(z)^T J_{g_i}(z).$$

This suggests several design principles:

- Encourage contraction in the symmetric part of the drift Jacobian.
- Control the magnitude and structure of diffusion Jacobians.
- Penalize positive eigenvalues of the combined drift–diffusion operator.

Thus, the spectral criterion offers a theoretically grounded approach to stability-aware regularization in continuous-time neural architectures.

7.6 Limitations and Scope

The proposed spectral condition is sufficient but not necessary. There exist stable systems that fail the identity-metric test but satisfy the general LMI condition with a nontrivial P. Furthermore, the nonlinear extension provides local mean-square stability and does not guarantee global stability without additional structural assumptions. Nevertheless, the simplicity and interpretability of the spectral criterion make it attractive for both theoretical analysis and practical neural system design.

7.7 Summary

The spectral drift–diffusion criterion refines classical scalar contraction tests by incorporating directional interactions between drift and diffusion. It occupies an intermediate position between scalar bounds and full LMI-based Lyapunov theory, offering a balance between analytical tractability and reduced conservativeness. This perspective highlights the importance of eigenstructure in stochastic neural stability and provides a foundation for further exploration of stability-aware neural dynamical modeling.

8 Limitations and Future Work

8.1 Scope of the Spectral Criterion

The spectral drift–diffusion condition developed in this work provides a sufficient criterion for mean-square exponential stability of linear multiplicative stochastic systems and a corresponding local condition for nonlinear stochastic neural ODEs. It is important to emphasize that the criterion is:

- **Sufficient but not necessary.**
- Restricted to **mean-square exponential stability**.
- Based on the **identity Lyapunov metric**.

Thus, while computationally simple and geometrically interpretable, the spectral test does not characterize the full stability region of the system.

8.2 Sufficiency vs. Necessity

For linear stochastic systems, the necessary and sufficient condition for mean-square exponential stability is the existence of a positive definite matrix $P \succ 0$ satisfying the Lyapunov matrix inequality

$$A^T P + PA + \sum_{i=1}^{m} B_i^T P B_i \prec 0.$$

Our spectral condition corresponds to the special case $P = I$. There exist stable systems for which no identity-metric certificate exists, but a nontrivial P can be found. Consequently, the spectral test may fail to certify stability even when the system is stable. Extending the spectral framework to characterize broader classes of admissible Lyapunov metrics remains an open challenge.

8.3 Mean-Square vs. Almost Sure Stability

The results of this paper concern mean-square exponential stability, defined by exponential decay of the second moment. Almost sure exponential stability, which concerns pathwise decay with probability one, is a stronger notion and generally requires different analytical techniques involving stochastic Lyapunov exponents and martingale convergence arguments. While mean-square stability often implies boundedness and robustness in practice, it does not guarantee almost sure exponential convergence. Establishing precise connections between the spectral criterion and almost sure stability remains an interesting direction for further investigation.

8.4 Identity Metric Restriction

The spectral condition evaluates the combined matrix

$$M = A + A^T + \sum B_i^T B_i$$

in the Euclidean metric. In contrast, the full LMI condition effectively searches over all quadratic metrics $V(z) = z^T P z$. Allowing general P introduces anisotropic contraction metrics that may significantly enlarge the certified stability region. However, computing such P requires solving semidefinite programs and sacrifices closed-form spectral interpretability. A promising research direction is to investigate structured or parameterized metric families that balance tractability with reduced conservativeness.

8.5 Additive Noise and Non-Vanishing Diffusion at Equilibrium

The present analysis focuses on multiplicative noise systems for which diffusion vanishes at equilibrium in the nonlinear extension. If diffusion does not vanish at equilibrium (additive noise), the origin cannot be mean-square exponentially stable in the strict sense, since noise continually injects energy into the system. In such cases, alternative notions of stability—such as bounded second moments, invariant distributions, or input-to-state stability—may be more appropriate. Extending the spectral perspective to such settings remains a nontrivial challenge.

8.6 Higher-Order Moments and Distributional Stability

The current framework analyzes second-moment behavior exclusively. In strongly nonlinear or heavy-tailed stochastic systems, higher-order moments may exhibit qualitatively different behavior. Stability of moments of order $p > 2$, or stability of full distributions, generally requires different techniques and may not be captured by quadratic Lyapunov functions. Developing spectral-type criteria for higher-order moment stability represents another open direction.

8.7 Future Research Directions

Several avenues for future work emerge naturally from this study:

Contraction Metrics for Stochastic Neural Systems. Extending the spectral condition to general contraction metrics may yield less conservative stability certificates and deeper connections to differential geometric contraction theory.

Data-Driven Stability Enforcement. In neural ODE training, one could penalize the maximal eigenvalue of the combined Jacobian matrix during optimization. Investigating practical algorithms for stability-aware learning remains an important direction.

Discrete-Time and Delayed Systems. Many neural architectures are implemented in discrete time or involve delay dynamics. Extending spectral drift–diffusion analysis to discrete-time stochastic systems or stochastic delay differential equations would broaden applicability.

Global Stability Conditions. The nonlinear extension presented here provides local mean-square stability. Establishing tractable sufficient conditions for global stochastic stability remains an open theoretical problem.

Connections to Random Dynamical Systems and Lyapunov Exponents. Relating the spectral criterion to sample-path Lyapunov exponents and multiplicative ergodic theory may offer deeper insight into long-term stochastic behavior.

8.8 Concluding Perspective

The spectral drift–diffusion criterion occupies an intermediate position between scalar contraction bounds and full LMI-based stochastic Lyapunov theory. While limited to identity-metric mean-square analysis, it provides a closed-form, interpretable, and computationally efficient stability test. Understanding how to generalize this spectral perspective to broader stochastic neural dynamical systems remains a promising and mathematically rich direction for future research.

9 Conclusion

In this work, we derived a spectral sufficient condition for mean-square exponential stability of linear multiplicative stochastic dynamical systems. The central result establishes that if

$$\lambda_{\max}\left(A + A^T + \sum_{i=1}^{m} B_i^T B_i\right) < 0,$$

then the equilibrium is mean-square exponentially stable, with an explicit decay rate determined by the maximal eigenvalue of the combined drift–diffusion matrix. Unlike classical scalar contraction–diffusion bounds, which impose isotropic constraints on drift and diffusion separately, the proposed spectral criterion captures directional interactions between these operators. This refined characterization allows stability certification in systems where scalar bounds fail, as demonstrated through an explicit counterexample and exact second-moment analysis. We further showed that the spectral condition corresponds to the identity-metric case of the general Lyapunov LMI framework, situating it between simple scalar tests and fully general quadratic stability theory. Extending the analysis to nonlinear stochastic neural ODEs via Jacobian linearization, we established a local spectral stability criterion based on the maximal eigenvalue of

$$J_f(z) + J_f(z)^T + \sum_i J_{g_i}(z)^T J_{g_i}(z).$$

This provides a practical and interpretable stability test for stochastic neural dynamical systems and suggests principled regularization strategies during model design and training. Overall, the spectral drift–diffusion framework offers a closed-form, computationally efficient, and geometrically

meaningful approach to stochastic stability analysis. By explicitly linking eigenstructure to second-moment dynamics, it provides both theoretical insight and practical guidance for the development of stable continuous-time neural architectures. I hope this perspective stimulates further research at the intersection of stochastic stability theory, neural dynamical systems, and stability-aware machine learning design.